

CHE213 Lab Report

Group 27

Cyclic Adsorption–Regeneration of KMnO_4 in a Packed Bed Column

1 Objectives

To study cyclic adsorption–regeneration behavior of KMnO_4 in a packed bed column and analyze the effect of regeneration on adsorption performance.

- Construct breakthrough curves
- Determine breakthrough and saturation times
- Calculate LUB and efficiency
- Compare fresh and regenerated beds

2 Materials

Activated charcoal, KMnO_4 solution, packed bed column, rotameter, pump, colorimeter, hot water source, measuring cylinders.

3 Experimental Setup

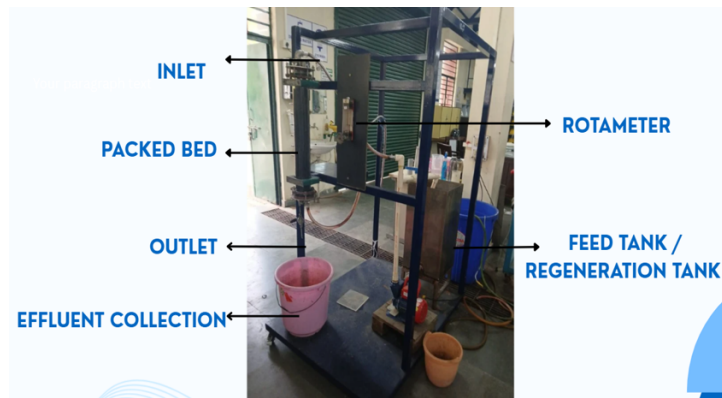


Figure 1: Packed Bed Adsorption Setup

4 Methodology

The experiment was conducted in two cycles to study adsorption–regeneration behavior in a packed bed column.

Cycle 1 (Adsorption – Fresh Bed): KMnO_4 solution was passed through a packed bed column containing activated charcoal at a constant flow rate. Effluent samples

were collected at regular time intervals, and absorbance was measured using a photocolorimeter.

Regeneration: After saturation, the feed was stopped and hot water (50–70°C) was passed through the column to regenerate the adsorbent.

Cycle 2 (Adsorption – Regenerated Bed): The adsorption experiment was repeated under identical conditions using the regenerated bed.

Data Processing: Absorbance values were converted to concentration using a calibration curve. Normalized concentration (C/C_0) was plotted against time to obtain breakthrough curves.

Controlled Variables: Flow rate, bed height, and inlet concentration.

Measured Variables: Time and absorbance.

5 Theory

Packed bed adsorption is used to remove solutes from a fluid by passing it through a column packed with an adsorbent. The performance is evaluated using a breakthrough curve, which plots normalized concentration versus time.

Normalized concentration is defined as:

$$\frac{C}{C_0}$$

where C is the outlet concentration and C_0 is the inlet concentration.

Due to deviation from Beer–Lambert law at higher concentrations, a quadratic calibration curve is used to convert absorbance (A) into concentration (C). This calibrated concentration is then used to compute C/C_0 .

Breakthrough time (θ_b) is defined as the time at which $C/C_0 = 0.1$.

Saturation time (θ_s) is defined as the time at which the outlet concentration approaches the inlet concentration ($C/C_0 \approx 1$).

If full saturation is not reached within the experimental duration, the final observed time is used as an approximation.

The Length of Unused Bed (LUB) is given by:

$$LUB = Z \left(\frac{\theta_s - \theta_b}{\theta_s} \right)$$

where Z is the bed height.

The adsorption efficiency is calculated as:

$$\eta = \left(1 - \frac{LUB}{Z} \right) \times 100$$

6 Equations

Beer–Lambert Law:

$$A = \epsilon l C$$

Normalized Concentration:

$$\frac{C}{C_0}$$

Note: Concentration C is obtained from absorbance using the calibration curve.
Concentration is obtained from absorbance using the quadratic calibration relation:

$$C = aA^2 + bA + c$$

Length of Unused Bed (LUB):

$$LUB = Z \left(\frac{\theta_s - \theta_b}{\theta_s} \right)$$

Efficiency:

$$\eta = \left(1 - \frac{LUB}{Z} \right) \times 100$$

Where: Z = bed height (m), θ_b = breakthrough time (min), θ_s = saturation time (min)

7 Tables

Table 1: Calibration Data for KMnO_4 (Absorbance vs Concentration)

KMnO_4	Water	Concentration (mol/L)	Absorbance (A)
0	10	0	0.00
1	9	3.615×10^{-4}	0.15
2	8	7.23×10^{-4}	0.21
3	7	1.0845×10^{-3}	0.27
4	6	1.446×10^{-3}	0.33
5	5	1.8075×10^{-3}	0.38
6	4	2.169×10^{-3}	0.42
7	3	2.5305×10^{-3}	0.46
8	2	2.892×10^{-3}	0.50
9	1	3.2535×10^{-3}	0.53
10	0	3.615×10^{-3}	0.55

Table 2: Breakthrough Data for Cycle 1 (Fresh Bed)

Time (min)	A	C (mol/L)	C/C ₀	Time (min)	A	C (mol/L)	C/C ₀
0	0.02	3.36×10^{-5}	0.0093	32	0.09	1.95×10^{-4}	0.0541
2	0.03	5.12×10^{-5}	0.0142	34	0.12	2.92×10^{-4}	0.0809
4	0.02	3.36×10^{-5}	0.0093	36	0.13	3.28×10^{-4}	0.0908
6	0.03	5.12×10^{-5}	0.0142	38	0.18	5.36×10^{-4}	0.1483
8	0.02	3.36×10^{-5}	0.0093	40	0.22	7.35×10^{-4}	0.2034
10	0.02	3.36×10^{-5}	0.0093	42	0.27	1.01×10^{-3}	0.2790
12	0.03	5.12×10^{-5}	0.0142	44	0.36	1.66×10^{-3}	0.4604
14	0.03	5.12×10^{-5}	0.0142	46	0.39	1.92×10^{-3}	0.5306
16	0.03	5.12×10^{-5}	0.0142	48	0.42	2.20×10^{-3}	0.6083
18	0.03	5.12×10^{-5}	0.0142	50	0.47	2.63×10^{-3}	0.7276
20	0.03	5.12×10^{-5}	0.0142	52	0.50	2.95×10^{-3}	0.8160
22	0.04	7.06×10^{-5}	0.0195	54	0.51	3.07×10^{-3}	0.8492
24	0.04	7.06×10^{-5}	0.0195	56	0.52	3.20×10^{-3}	0.8851
26	0.05	9.19×10^{-5}	0.0254	58	0.51	3.07×10^{-3}	0.8492
28	0.06	1.15×10^{-4}	0.0318	60	0.52	3.20×10^{-3}	0.8851
30	0.07	1.40×10^{-4}	0.0387				

Table 3: Breakthrough Data for Cycle 2 (Regenerated Bed)

Time (min)	A	C (mol/L)	C/C ₀	Time (min)	A	C (mol/L)	C/C ₀
0	0.03	5.12×10^{-5}	0.0142	32	0.21	6.83×10^{-4}	0.1888
2	0.03	5.12×10^{-5}	0.0142	34	0.24	8.46×10^{-4}	0.2340
4	0.03	5.12×10^{-5}	0.0142	36	0.28	1.10×10^{-3}	0.3013
6	0.03	5.12×10^{-5}	0.0142	38	0.35	1.59×10^{-3}	0.4387
8	0.03	5.12×10^{-5}	0.0142	40	0.39	1.91×10^{-3}	0.5284
10	0.04	7.06×10^{-5}	0.0195	42	0.44	2.36×10^{-3}	0.6520
12	0.04	7.06×10^{-5}	0.0195	44	0.48	2.74×10^{-3}	0.7600
14	0.05	9.19×10^{-5}	0.0254	46	0.50	2.95×10^{-3}	0.8171
16	0.05	9.19×10^{-5}	0.0254	48	0.52	3.20×10^{-3}	0.8762
18	0.06	1.15×10^{-4}	0.0318	50	0.53	3.28×10^{-3}	0.9066
20	0.07	1.40×10^{-4}	0.0387	52	0.54	3.39×10^{-3}	0.9374
22	0.08	1.67×10^{-4}	0.0461	54	0.53	3.28×10^{-3}	0.9066
24	0.09	1.95×10^{-4}	0.0541	56	0.54	3.39×10^{-3}	0.9374
26	0.11	2.58×10^{-4}	0.0714				
28	0.13	3.28×10^{-4}	0.0908				
30	0.15	4.06×10^{-4}	0.1123				

8 Plots

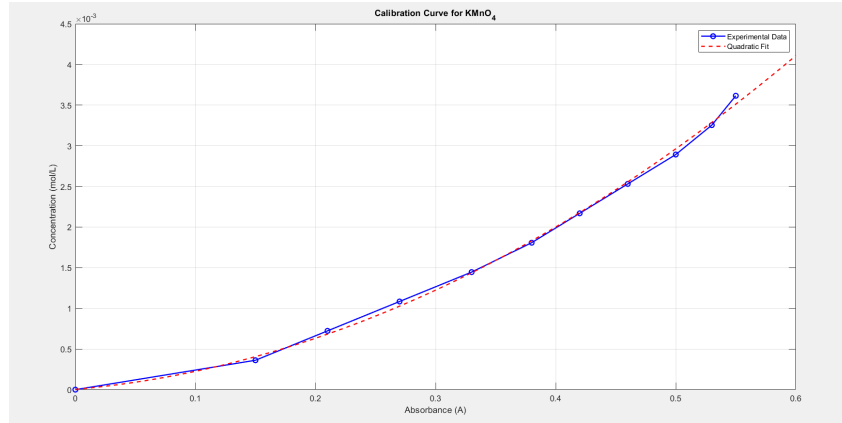


Figure 2: Calibration Curve for KMnO_4 (Absorbance vs Concentration)

Observation: The calibration curve shows slight deviation from linearity, hence a quadratic fit is used for accurate concentration estimation.

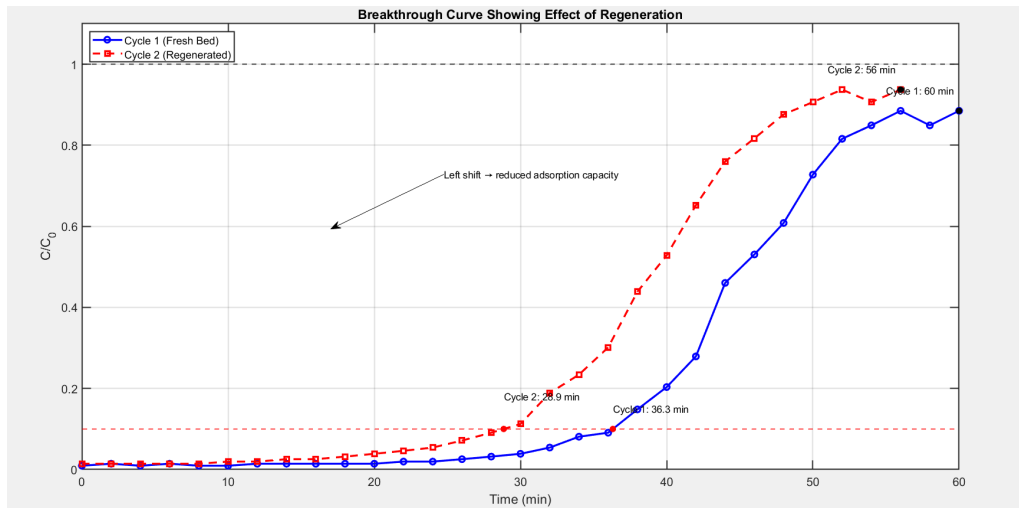


Figure 3: Breakthrough Curves for Cycle 1 (Fresh Bed) and Cycle 2 (Regenerated Bed)

Observation: The breakthrough curve for Cycle 2 shifts to the left compared to Cycle 1, indicating earlier breakthrough and reduced adsorption capacity after regeneration.

9 Calculations

Calibration Curve

At higher concentration, Beer–Lambert law deviates due to non-ideal solution behavior and instrument limitations. Hence, a quadratic fit is used:

$$C = aA^2 + bA + c$$

From MATLAB fitting:

$$C = 9.2 \times 10^{-3} A^2 + 1.3 \times 10^{-3} A + 3.9 \times 10^{-6}$$

Sample Calculation:

For $A = 0.30$,

$$C = 9.2 \times 10^{-3}(0.30)^2 + 1.3 \times 10^{-3}(0.30) + 3.9 \times 10^{-6}$$

$$C \approx 1.22 \times 10^{-3} \text{ mol/L}$$

This equation is used to convert absorbance values into concentration.

Cycle 1 (Fresh Bed)

Given: $\theta_b = 36.3 \text{ min}$, $\theta_s = 60 \text{ min}^*$, $Z = 1 \text{ m}$

$$LUB = Z \left(\frac{\theta_s - \theta_b}{\theta_s} \right) = 1 \times \frac{60 - 36.3}{60} = 0.395 \text{ m}$$

$$\eta = \left(1 - \frac{LUB}{Z} \right) \times 100 = 60.5\%$$

**Saturation not fully reached; final experimental time used as approximation.*

Cycle 2 (Regenerated Bed)

Given: $\theta_b = 28.9 \text{ min}$, $\theta_s = 56 \text{ min}$, $Z = 1 \text{ m}$

$$LUB = 1 \times \frac{56 - 28.9}{56} = 0.484 \text{ m}$$

$$\eta = (1 - 0.484) \times 100 = 51.6\%$$

10 Results and Discussion

The breakthrough curves for Cycle 1 (fresh bed) and Cycle 2 (regenerated bed) are shown in Figure 3. A leftward shift of the curve for Cycle 2 is observed, indicating earlier breakthrough and reduced adsorption performance.

For Cycle 1, the breakthrough time is 36.3 min and the saturation time is 60 min*. For Cycle 2, these values decrease to 28.9 min and 56 min, respectively. This indicates earlier breakthrough and reduced adsorption capacity in the regenerated bed. (*Saturation not fully reached; final experimental time used as approximation.)

The Length of Unused Bed (LUB) increases from 0.395 m (Cycle 1) to 0.484 m (Cycle 2), indicating poorer utilization of the adsorbent bed after regeneration. This suggests expansion of the mass transfer zone due to increased resistance to mass transfer.

The adsorption efficiency decreases from 60.5% in Cycle 1 to 51.6% in Cycle 2, confirming a reduction in adsorption capacity after regeneration.

The broader breakthrough curve observed in Cycle 2 indicates a wider mass transfer zone, which implies reduced mass transfer efficiency and higher intraparticle diffusion resistance.

These effects can be attributed to incomplete regeneration, leading to:

- Residual solute within pores
- Blockage of active adsorption sites
- Increased resistance to mass transfer

Performance Comparison

$$\text{Breakthrough time decrease} = \frac{36.3 - 28.9}{36.3} \times 100 = 20.4\% \approx 20\%$$

$$\text{Efficiency decrease} = \frac{60.5 - 51.6}{60.5} \times 100 = 14.7\% \approx 15\%$$

$$\text{LUB increase} = \frac{0.484 - 0.395}{0.395} \times 100 = 22.5\% \approx 23\%$$

11 Sources of Error

- Flow rate fluctuations affecting residence time
- Instrumental errors in absorbance measurement
- Air bubbles or improper packing causing channeling
- Incomplete regeneration of adsorbent
- Temperature variations during regeneration

12 Physical Interpretation

As observed from Figure 3, the regenerated bed shows earlier breakthrough, indicating reduced adsorption capacity of the adsorbent.

This occurs due to incomplete regeneration, where some adsorption sites remain occupied by residual solute.

The increase in LUB signifies inefficient utilization of the bed length, while a broader mass transfer zone (MTZ) indicates reduced mass transfer efficiency and higher intraparticle diffusion resistance.

Overall, regeneration leads to degradation in adsorption performance over successive cycles.

13 Industrial Relevance

Packed bed adsorption systems are widely used in water and wastewater treatment for removal of dyes and contaminants, as well as in gas purification processes.

Regeneration allows reuse of adsorbents, thereby reducing operational costs. However, as observed in this study, regeneration leads to reduced efficiency and earlier breakthrough.

This highlights the importance of optimizing regeneration techniques to maintain adsorption performance.

Efficient regeneration is essential for sustainable and cost-effective industrial adsorption systems.

14 Conclusion

The experiment demonstrates that adsorption efficiency decreases after regeneration due to incomplete recovery of active sites.

The regenerated bed shows earlier breakthrough, increased LUB, and reduced efficiency compared to the fresh bed, as seen in Figure 3.

These results highlight the importance of optimizing regeneration processes to improve adsorption performance and ensure sustainability in industrial applications.